# DWI harmonization & cross-validation: audit before new experiments

Finding: the current end-to-end evaluation is not leakage-free.
Global ComBat precedes subject-level cross-validation.

Existing safeguards
- Participant-grouped outer CV
- Participant-grouped early-stopping split
- Training-only target age scaling

These protect model fitting, but not upstream ComBat.

Required changes
- Fit learned preprocessing inside each split
- Select topology / architecture inside outer training
- Define allowed harmonization covariates
- Cross-fit clinical reference predictions

# Current implementation: global preprocessing comes first

Raw DWI
4 cohorts → Global edge QC
+ log1p → Joint ComBat
AGE / SEX /
optional DX → Saved harmonized
84 × 84 matrices

Then: CN GroupKFold → grouped 80/20 early stopping → outer OOF predictions

Repeat for five graph thresholds → select by pooled OOF MAE → report selected OOF score

Evidence: run_DWI_harmonization.py:2226–2253, 2542–2675; GNN_DWI_BAG.py:710–753, 940–1016

# Audit findings: confirmed code behavior and its implications

| | |
|---|---|
| Global ComBat | All retained samples enter neuroCombat before CV. Held-out distributions influence the representation. [source 2638; GNN:46] |
| Global feature mask | Finite / nonconstant edge mask uses all rows. Learn data-dependent masks on training rows. [source 2226–2253] |
| Threshold selection | Five candidates ranked on pooled outer OOF MAE. Selected score is a model-selection estimate. [source GNN:940–1016] |
| Covariates | True AGE always preserved; GROUP conditional on completeness/rank. Define age-conditioned use and diagnosis circularity. [source 2416–2675] |
| Participant grouping | Only terminal _Y<number> removed. Grouping exists; actual visit IDs still require a data audit. [source GNN:145–155] |

# Proposed pipeline: split first; fit within the split

Freeze eligible sample IDs + canonical participant groups + outer folds

OUTER TRAINING ONLY

1. Inner grouped train / validation splits
2. Fit edge QC + harmonization on inner train
3. Transform inner validation with frozen state
4. Select topology, capacity, epoch budget
5. Refit on all outer training with frozen choices

OUTER TEST: LOCKED

- Never used for preprocessing fit or selection
- Apply outer-training transform
- Predict once per chosen model
- Save raw age predictions and fold IDs
- Aggregate OOF metrics after all folds

Gate: verify a train/apply harmonizer that preserves required covariate effects; never concatenate test rows to refit.

# Define the scientific target before choosing harmonization

Known-site, new-participant prediction
- Group by participant in every split
- Ensure training support for each test batch
- Learn correction only from training rows
- Predefine unsupported-batch handling

Age-conditioned analysis
True AGE may be available, but results must explicitly describe that conditioning.

Unseen-site generalization
- Hold out a site / cohort explicitly
- Standard known-batch parameters do not resolve an unseen batch
- Any target-site adaptation needs its own declared evaluation design

Clinical diagnosis analyses
Avoid preserving diagnosis in the primary transform when later testing diagnosis associations.

# cBAG: separate reference calibration from evaluated subjects

Current: fit BAG ~ AGE on CN OOF predictions, then subtract that trend from final-model BAG.
CN final-model predictions are in-sample; existing cBAG is not fully cross-fitted.

Proposed: within each outer training set, obtain inner OOF CN predictions and estimate the age-bias trend.
Freeze the trend and apply it to the outer-held-out BAG. Keep all visits of a subject together.
For clinical comparisons, use held-out predictions for all evaluated subjects or a prespecified independent clinical set.

cBAG = predicted age − chronological age − (training-reference slope × age + intercept)

Evidence: GNN_DWI_BAG.py:1140–1188. Clinical design remains provisional pending variable inventory.

# Next steps: controlled capacity study and clinical inventory

Learning curves / capacity
Current: 3 GINE layers, width 64, dropout 0.30
Proposed: 2 × 32; 3 × 32; current 3 × 64
Log epoch-level train + validation MAE and loss, best epoch, parameters and runtime.
Use eval-mode training MAE for comparable curves.
Select configurations inside outer training.
Current fit_one_fold does not persist full curves.

Clinical inventory / cBAG
Inventory ADNI, HABS, ADRC, ADDecode separately: definitions, units, missingness, visit linkage and overlap.
Verify HABS CN override from diagnosis evidence.
Predefine primary associations and covariates after inventory.

This deck is a source audit and proposal.
No corrected performance claims or new experiments yet.